

Indian Institute of Information Technology Dharwad

Department of ECE and Data Science

Tensor Methods for Multimodal Data Analysis and AI

NeuroTensor

Graph-Regularized Multi-Way EEG Tensor Decomposition

for Differential Classification of Alzheimer's Disease,
Frontotemporal Dementia, and Cognitively Normal Controls

Comprehensive Technical Report
and Reproducibility Record

OpenNeuro dataset ds004504, local version 1.0.9

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Report Basis and Scope

This report is a faithful consolidation of the source dataset, preprocessing code, versioned experiment protocols, saved machine-readable results, and verification artifacts present in the project workspace on 6 September 2026. The original proposal and the course slides supplied by Dr. Sibasankar Padhy are treated as contextual academic sources, not as instructions to claim work that was not run.

The five-member table on the title page and the proposal-assigned responsibilities in Chapter 1 are transcribed from the supplied formal proposal. They record the planned module allocation; they are not used to infer undocumented individual contribution percentages.

All reported predictive results are subject-level cross-validation estimates from one public cohort. They are not external validation, medical-device evidence, or clinical diagnostic performance. The strongest observed result, 59.7% mean balanced accuracy for the tuned RBF SVM family, is explicitly secondary to the predeclared V4 automatic-selection endpoint of 54.5%.

Integrity note

No source BIDS recording, earlier processed partition, model artifact, or prior result was deleted or silently overwritten during the project. Corrections to electrode order and boundary handling are documented as corrections, and earlier outputs are retained as historical exploratory evidence.

Executive Summary

This project investigates whether multi-way representations of resting-state EEG can support three-class subject-level discrimination among Alzheimer’s disease (AD), frontotemporal dementia (FTD), and cognitively normal controls (CN). The source is the public OpenNeuro ds004504 cohort: 88 people, 19 scalp channels, 500 Hz eyes-closed recordings, and author-supplied artifact- cleaned derivatives [Miltiadous et al., 2023, OpenNeuro, 2026].

The work evolved through four experiments. It began with a deliberately balanced 69-person primary cohort and exploratory non-negative spectral tensor methods. It then corrected channel labels and EEGLAB boundary handling, introduced training-only TT/Tucker bases and nested subject cross-validation, expanded to all 88 people using raw-waveform delay moments plus class/subject weighting, and finally performed a bounded 139-configuration comparison of logistic regression, RBF SVM, random forest, XGBoost, and supervised TT variants.

Headline outcome

The strongest observed classifier-family result is the tuned RBF SVM at **59.7% mean balanced accuracy** (repeat SD 0.9 percentage points), with 57.4% AD recall, 43.5% FTD recall, and 78.2% CN recall. The V4 predeclared complete-pipeline endpoint is **54.5%**. This is useful course evidence above the 33.3% single-class balanced-accuracy baseline, but it is not an independently confirmed 60% clinical result.

The central empirical lesson is that tensorization alone did not guarantee a gain. Corrected V2 Tucker/HOSVD reached 56.0% as the strongest method family, while V3’s same-cohort spectral control (55.8%) exceeded every waveform-only branch (49.9–51.8%). V4’s TT/spectral fusion family reached 59.1%, but training-only feature selection could favor spectral predictors, so the result does not isolate a causal TT contribution. Across versions, FTD remains the hardest class and the principal data limitation is 23 independent FTD people, not merely the number of four-second windows.

The pipeline is methodologically stronger than its first prototype because all current learned stages are fitted on training people only; windows from a person never cross validation boundaries; per-person pooling prevents long recordings from becoming pseudo-replicates; class weights equalize diagnosis-level loss; and verification replays saved predictions, checks split membership, confirms training-only bases/scalers, and fingerprints source arrays.

Figure 1 summarizes how these stages connect.

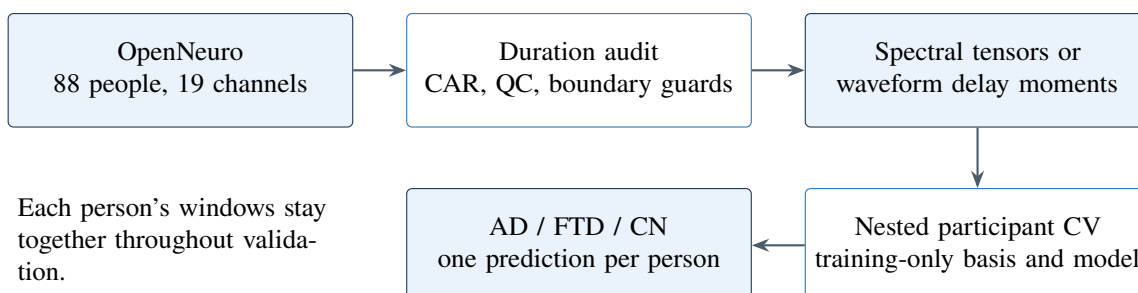


Figure 1: Completed NeuroTensor workflow. Predictive evaluation is performed at the participant level.

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1 Problem Definition, Course Alignment, and Scope

1.1 Clinical and computational motivation

Alzheimer’s disease and frontotemporal dementia can present overlapping cognitive and behavioral symptoms, while their electrophysiological signatures may vary across people, time, frequency, and scalp location. Resting-state EEG is non-invasive and comparatively accessible, but its high temporal resolution creates a modeling problem: a single recording is not naturally a flat table. It has participant, channel, time, frequency, and potentially connectivity modes. The course premise that physiological data can be organized as *channels* $\times$ *time* $\times$ *frequency* $\times$ *subject* therefore maps directly to this dataset [Padhy, 2026a].

The project asks a concrete question:

Can carefully tensorized, low-rank representations of artifact-cleaned, eyes-closed resting EEG support reproducible subject-level classification of AD, FTD, and CN while preserving multi-way structure and avoiding window-level leakage?

The answer is investigated comparatively rather than assumed. Matrix baselines, non-negative CP/Tucker, Tensor Train, waveform-delay Tucker/TT, a supervised TT variant, spectral controls, and several classifier families are all retained in the record. This makes the project as much an evaluation of representation choices as a search for a strong classifier.

1.2 Course learning outcomes addressed

The completed work operationalizes the course concepts in four ways:

1. **Tensorization as a modeling decision.** The same EEG source is represented as a subject-channel-frequency-time tensor, a signed channel-frequency-local-segment tensor, and a channel-lag-position delay tensor. These are not cosmetic reshapes; each encodes a different invariance.
2. **Low-rank structure.** CP, Tucker/HOSVD, and TT decompositions replace large mode products with compact factors or reusable bases, following the rank-one and multilinear ideas emphasized in Module 2 [Padhy, 2026b, Kolda and Bader, 2009, Oseledets, 2011].
3. **Multimodal interactions.** Channel, frequency, local time, lag, and participant modes are kept explicit long enough for decomposition to exploit their interactions.
4. **Responsible model comparison.** The project demonstrates why tensor structure, optimization convergence, and prediction accuracy are distinct questions. Strong reconstruction or a converged optimizer does not guarantee strong held-out clinical classification.

1.3 Proposal-to-implementation traceability

The one-page proposal described an ambitious graph-regularized tensor biomarker system. Table 1.1 distinguishes completed work from planned but unimplemented extensions. This prevents proposal language from being mistaken for results.

Table 1.1: Traceability from the formal proposal to the implemented workspace.

Proposal element	Implemented counterpart	Status and interpretation
Subject $\times$ channel $\times$ frequency $\times$ time tensor	V1 tensor of shape $69 \times 19 \times 59 \times 10$ and V2 tensors of shape $19 \times 59 \times 3$ per window	Completed, with Welch/spectrogram PSD rather than Morlet wavelets.

Proposal element	Implemented counterpart	Status and interpretation
Dynamic phase-locking value connectivity	No dPLV feature branch in the saved experiments	Not implemented; no connectivity-result claim is made.
Graph-regularized non-negative CP/Tucker	Preliminary graph NCP in V1; corrected fixed anatomical pre-smoothing before TT in V2	Partially implemented. The old graph-NCP objective had a scale-consistency caveat and remains exploratory.
CORCONDIA and uniqueness analysis	Rank sweeps, reconstruction diagnostics, orthogonality tests, and numerical TT checks	CORCONDIA itself was not implemented.
LOSO-CV	Repeated five-fold subject CV in V1 and nested repeated five-fold subject CV in V2–V4	Replaced with repeated nested subject CV to make bounded model/rank selection feasible.
PSD and Riemannian baselines	PSD/spectral controls in every relevant stage	PSD completed; Riemannian covariance baseline not implemented.
MMSE correlation and biomarker topoplots	MMSE retained in metadata but excluded from predictors; no validated factor-MMSE analysis	Not implemented; this report does not imply clinical severity biomarkers.
Three-way AD/FTD/CN classifier	Logistic regression, SVM, histogram boosting, random forest, XGBoost, and supervised TT	Completed as subject-level comparative experiments.

1.4 Team and proposal-assigned modules

The formal proposal assigned the following course modules. Wording here is normalized for the report, while the names, roll numbers, and allocation follow the supplied proposal.

Member	Roll number	Proposal-assigned technical module
Neeruganti Jeevan Sai	24bds046	Multi-way tensor construction, time-frequency representation, synchrony-array design, and normalization.
Kothapalli Tharun Chowdary	24bds032	Optimization and regularization, graph-aware CP/Tucker design, and electrode adjacency construction.
Vatipalli Abhiram	24bds087	Rank selection and stability, convergence analysis, hyperparameter exploration, and residual profiling.
Palyam Maruthi	24bcs098	Biomarker mapping and statistics, group-level comparison, topographic projection, and MMSE association planning.
Shaik Moiz	24bcs133	Classification and benchmarking with SVM/logistic models, metric analysis, and baseline comparison.

Table 1.2: Team list and planned module allocation from the formal proposal.

1.5 Research questions and success criteria

The completed pipeline addresses five research questions:

1. Which diagnosis limits balanced participant and duration coverage?
2. Can a reproducible primary cohort be built without discarding excluded windows or synthetically oversampling FTD?
3. Which tensorization preserves enough discriminative structure for subject-level prediction?

4. Do TT/Tucker representations improve over spectral controls when every learned step is confined to training people?
5. How much does bounded classifier and optimization tuning improve the complete procedure, and which diagnosis remains difficult?

Success is not defined as reaching an arbitrary 65–70% threshold. A credible course result must instead be reproducible, leakage-aware, explicit about the primary endpoint, complete about negative findings, and honest about the lack of external validation.

2 Dataset Description and Initial Risk Analysis

2.1 Source and acquisition

The project uses OpenNeuro dataset ds004504, local version 1.0.9, released under CC0. It contains routine eyes-closed resting-state recordings from 88 people: 36 AD, 23 FTD, and 29 CN. Data were acquired with a clinical Nihon Kohden system at 500 Hz using 19 scalp electrodes positioned according to the international 10–20 system. The accompanying dataset paper describes the cohort, recording protocol, BIDS structure, MMSE metadata, and author-supplied preprocessing [Miltiadous et al., 2023, OpenNeuro, 2026].

The local derivative EEGLAB files are used because the dataset authors recommend their preprocessed release. Those derivatives include approximately 0.5–45 Hz band limitation, mastoid referencing, Artifact Subspace Reconstruction (ASR), ICA, and automatic ICLabel-based component removal. ICLabel is an automated classifier for EEG independent components [Pion-Tonachini et al., 2019]. The project therefore does not repeat ASR or ICA and does not relabel the derivatives as raw unprocessed recordings.

2.2 Participants and demographics

Table 2.1 reports the local participant inventory. Age and sex are used to audit selection, not as predictive features. MMSE is retained as metadata but excluded from all classifiers, preventing an outcome- adjacent clinical score from inflating EEG-only performance.

Group	People	Female	Male	Age range	Age mean
AD	36	24	12	49–79	66.39
FTD	23	9	14	44–78	63.65
CN	29	11	18	57–78	67.90
Total	88	44	44	44–79	–

Table 2.1: Source participant distribution. Sex coding follows the supplied metadata.

2.3 Recording-duration coverage

The source contains 70,590.3 seconds (19.61 hours) of recording. The duration imbalance is substantial: FTD contributes both the fewest people and the least total time. This makes FTD the correct reference margin for a controlled three-class primary cohort.

Across all participants, the longest recording is sub-010 (AD, 1,291.1 s) and the shortest is sub-003 (AD, 307.1 s). Within FTD, the range is sub-070 at 479.1 s to sub-074 at 1,026.9 s; within CN, it is sub-060 at 751.5 s to sub-040 at 1,017.1 s. Non-destructive duration-ranked views were created under preprocessing/01_duration_sorted; the source files themselves were not moved or renamed.

Group	People	Seconds	Hours	Complete 4-s windows
AD	36	29,403.3	8.17	7,267
FTD	23	16,753.4	4.65	4,137
CN	29	24,433.6	6.79	6,015
Total	88	70,590.3	19.61	17,419

Table 2.2: Source duration and complete non-overlapping four-second window inventory.

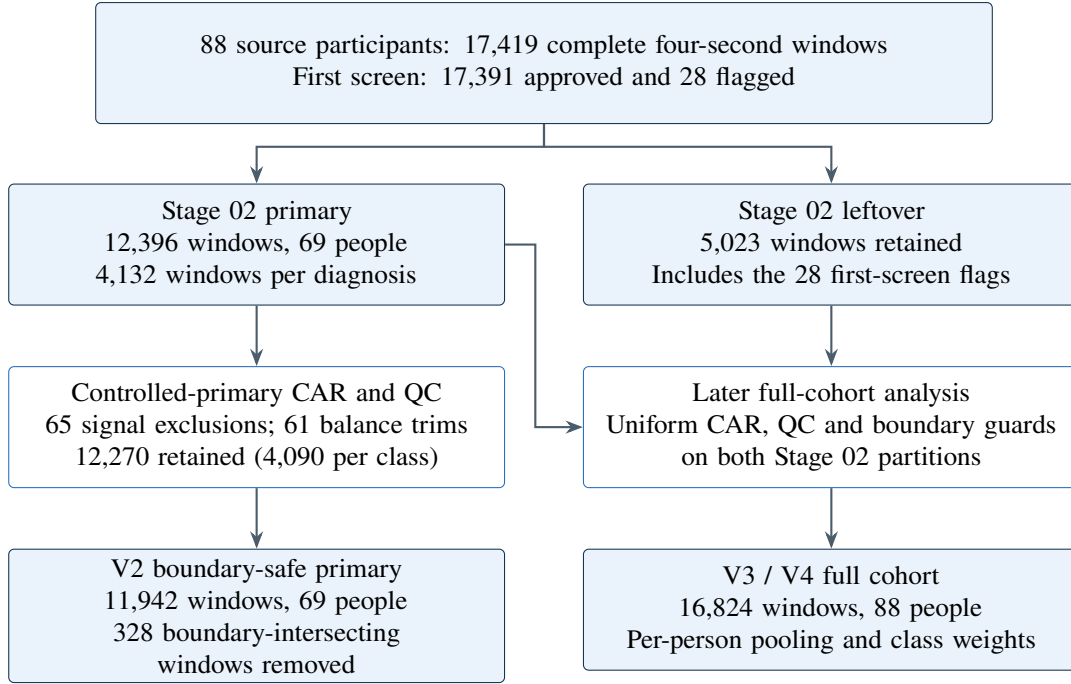


Figure 2.1: Dataset lineage. The controlled-primary path and the later full-cohort path create separate analysis indexes. Both Stage 02 partitions are preserved.

2.4 Initial methodological risks

2.4.1 Class count versus window count

The 17,419 complete windows are not 17,419 independent patients. Windows from one recording share anatomy, acquisition conditions, diagnosis, and temporal context. A random window split would permit near-duplicate subject signatures to appear in both training and test sets. Every valid experiment must therefore split by subject and pool predictions or features at the subject level.

2.4.2 Duration confounding

If all windows are used naively, AD supplies 1.76 times as many complete windows as FTD. A classifier may then learn recording exposure or obtain an overly stable estimate for long recordings. The controlled primary cohort uses duration buckets and equal windows; the later full cohort averages windows within person and applies diagnosis-level sample weights.

2.4.3 Demographic mismatch

FTD is younger on average and has a different sex ratio. Exact demographic identity is neither feasible nor desirable with 23 people per class. The design uses broad matching for the controlled cohort, records remaining differences, and excludes age and sex from model predictors.

2.4.4 Reference and channel-order dependence

Graph methods are sensitive to electrode labels and coordinate order. The first tensorizer used a hard-coded order that did not match the EEGLAB row order. Later code reads and verifies `chanlocs` for every subject. This correction changes anatomical interpretation, not channel-permutation-invariant operations such as PCA, global statistics, or common-average reference.

2.4.5 Boundary discontinuities

EEGLAB boundary events indicate data cuts. Four-second windows near such events can contain artificial discontinuities even after amplitude QC. V2 and later analyses therefore remove any window intersecting a fixed 0.5 s guard on either side of each boundary.

3 Preprocessing, Cohort Construction, and Signal Quality Control

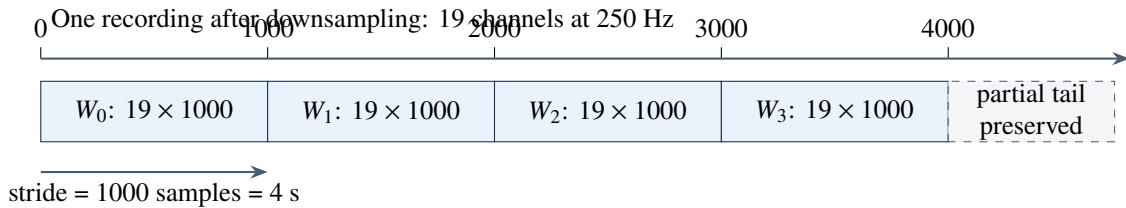
3.1 Design principles

The preprocessing pipeline follows four constraints: preserve source data, create a controlled primary dataset, retain every excluded complete window in a separate partition, and make all selection rules reproducible. Figure 2.1 summarizes the counts; this chapter gives the algorithmic details.

3.2 Window formation and first-pass QC

Each derivative recording is loaded from its EEGLAB `.set` file and checked for 19 channels at 500 Hz. Because the supplied derivative is already low-pass filtered at 45 Hz, retaining every second sample gives 250 Hz without aliasing the retained band. Each record is split into non-overlapping four-second arrays of shape 19×1000 . Per-channel temporal means are removed. The incomplete terminal segment from each of the 88 recordings is preserved separately; the 88 tails total 38,780 samples per channel or 155.12 seconds.

Figure 3.1 shows the sample indexing, stride, and separate handling of incomplete tails. Here N is the downsampled recording length; these four-second windows are distinct from the overlapping local segments used later inside spectral and waveform-delay tensorization.



$$W_k = X[:, 1000k : 1000(k + 1)], k = 0, \dots, \lfloor N/1000 \rfloor - 1$$

No overlap between four-second windows. QC and duration-based selection act on complete windows. Later analysis rejects any window intersecting a boundary guard $[b - 0.5, b + 0.5]$ seconds.

Figure 3.1: Segmentation and window generation, using a schematic recording with four complete windows and a partial tail. Indices are zero-based and slices exclude the right endpoint.

The first label-blind screen flags non-finite values, a channel standard deviation below 0.1 microvolt, or maximum absolute amplitude above 500 microvolts. It approves 7,252 AD, 4,132 FTD, and 6,007 CN windows. Flagged windows remain in the leftover partition.

3.3 FTD-referenced duration buckets

The 23 FTD participants are sorted by duration and divided by rank into 8 short, 7 medium, and 8 long participants. Midpoints between adjacent boundary durations define thresholds shared with the other

classes:

$$\text{bucket}(d) = \begin{cases} \text{short}, & d \leq 647.60 \text{ s}, \\ \text{medium}, & 647.60 < d \leq 806.45 \text{ s}, \\ \text{long}, & d > 806.45 \text{ s}. \end{cases} \quad (3.1)$$

The FTD window targets are 1,160 short, 1,222 medium, and 1,750 long, totaling 4,132.

3.4 Participant matching as a mixed-integer assignment

The function `match_group()` formulates a binary assignment between each FTD reference person r and candidate c from AD or CN. With $x_{rc} \in \{0, 1\}$, the cost is

$$J_{rc} = 1000 \max(0, w_r - w_c) + 300\mathbb{I}(s_r \neq s_c) \quad (3.2)$$

$$+ 10|a_r - a_c| + 0.1|w_r - w_c|, \quad (3.3)$$

where w is the count of approved windows, s is sex, and a is age. The mixed-integer linear program minimizes $\sum_{r,c} J_{rc}x_{rc}$ subject to one candidate per reference, at most one reference per candidate, fixed feasible male quotas (12 for AD, 13 for CN), and sufficient total window capacity inside each duration bucket. The code uses SciPy’s `milp()` with a 120 s limit. It does not duplicate a participant.

If a selected recording has more approved windows than assigned, the function `even_temporal_selection()` spreads chosen windows over the full recording. For m available windows and target k , the selected zero-based positions are

$$p_j = \left\lfloor \frac{(j + 0.5)m}{k} \right\rfloor, \quad j = 0, \dots, k - 1. \quad (3.4)$$

This avoids taking only the beginning of longer recordings. The resulting first primary partition contains 4,132 windows per class, 23 participants per class, and 49,584 seconds. Its broadly matched demographics are shown in Table 3.1.

Group	People	F	M	Age mean (SD)	Age range	Windows
AD	23	11	12	65.74 (7.35)	49–79	4,132
FTD	23	9	14	63.65 (8.22)	44–78	4,132
CN	23	10	13	66.87 (5.24)	57–78	4,132

Table 3.1: Participant composition of the first controlled primary cohort.

The first leftover dataset contains 5,023 complete windows: 3,135 AD, 5 FTD, and 1,883 CN. Of these, 4,995 pass the first conservative screen and 28 carry initial extreme-amplitude flags. Reasons are explicit: unselected participant, excess approved window, or QC flag. The primary and leftover IDs are disjoint and jointly exhaust all 17,419 complete source windows.

3.5 Signal audit and common-average reference

The audit computes amplitude, adjacent-sample step, channel standard deviation, line-noise-band power, band powers, entropy, and across-channel common-variance statistics. The first primary set contains no material non-finite, repeated- value, or residual 50 Hz issue. Across-channel common variance is high (median 0.953; 95th percentile 0.986), but this alone is not evidence of electrode bridging or artifact because scalp volume conduction and a shared mastoid reference naturally induce correlation.

The project applies common-average reference (CAR) at each time sample:

$$x'_{c,t} = x_{c,t} - \frac{1}{19} \sum_{j=1}^{19} x_{j,t}, \quad (3.5)$$

followed by removal of each channel’s temporal mean inside the four-second window. No additional broad filter, ASR, ICA, or 50 Hz notch is applied.

3.6 Fixed post-CAR exclusion rules

The cleaned controlled dataset uses the following label-blind thresholds:

- maximum absolute amplitude at least 400 microvolts;
- maximum adjacent-sample step at least 150 microvolts per 4 ms;
- a separately recorded extreme-step category at least 300 microvolts;
- minimum channel standard deviation below 1 microvolt;
- non-finite values.

Sixty-five windows fail at least one post-CAR rule: 9 AD, 42 FTD, and 14 CN. Reason totals overlap because one window may trigger multiple conditions: 52 ordinary abrupt steps, 9 extreme abrupt steps, 8 extreme amplitudes, and 1 near-flat channel. Most suspicious FTD windows are concentrated in sub-086; the participant remains in the cohort. High 30–45 Hz activity in sub-067, sub-077, and sub-085 is documented but is not used as a class-sensitive exclusion criterion.

After QC, FTD has 4,090 windows and again limits the equal-window primary cohort. The pipeline moves 33 AD and 28 CN medium-duration windows to an excluded partition using a deterministic quality-first trim. The final controlled primary therefore has 12,270 windows, exactly 4,090 per diagnosis, 69 participants, and 49,080 seconds. The excluded companion contains 126 windows, exactly 42 per diagnosis: 65 signal-QC exclusions plus 61 balance-only trims. The shuffle seeds are 4506 for the retained primary and 4507 for the excluded partition.

3.7 Boundary-safe V2 index

The V2 preparation code reads the actual channel labels and coordinates from every EEGLAB header and asserts a common row order:

Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, Fz, Cz, Pz.

It then removes a primary window if its $[t, t + 4)$ interval intersects any EEGLAB boundary time expanded by 0.5 s on both sides. This removes 328 analysis windows and retains 11,942: 3,985 AD, 4,000 FTD, and 3,957 CN. The stored clean-primary arrays remain unchanged; V2 uses a separate index.

3.8 Full-cohort V3/V4 analysis index

For the approved full-cohort experiment, the code reads both original stage-02 primary and leftover arrays, then reapplies CAR, the same thresholds, and the boundary guard uniformly. It retains 16,824 windows from all 88 people: 7,030 AD, 4,000 FTD, and 5,794 CN. Of these, 4,828 originate from the old leftover partition, and 54 are recovered balance-only trims. All 11,942 V2 windows remain included, no formerly artifact-excluded primary window is reintroduced, and 595 excluded IDs retain their reasons.

This design uses more signal without pretending to create additional FTD participants. Later models average windows within person and weight the 23 FTD participants as a class rather than oversampling their windows.

4 Tensorization and Low-Rank Decomposition Foundations

4.1 Why tensorize EEG?

A tensor is a multi-way array $\mathcal{X} \in \mathbb{R}^{I_1 \times \dots \times I_N}$. The course materials distinguish order from dimension and emphasize that flattening can obscure interactions among modes [Padhy, 2026a]. For EEG, a value can be indexed simultaneously by participant, channel, frequency, and local time. Vectorizing these axes is always algebraically possible, but it removes the explicit statement that neighboring frequencies, electrodes, and time bins have different meanings.

Tensorization is therefore a hypothesis about structure. In this project:

- a frequency tensor assumes subject patterns can be expressed through separable or multilinear channel-frequency-time factors;
- a local-segment logPSD tensor preserves signed deviations from a training mean and supports orthogonal TT/Tucker projection;
- a waveform-delay tensor assumes short-lag covariance contains useful information even when absolute waveform phase is not aligned across people.

4.2 Rank-one structure and CP decomposition

A third-order rank-one tensor is the outer product $\mathbf{a} \circ \mathbf{b} \circ \mathbf{c}$ with entries $x_{ijk} = a_i b_j c_k$. CP represents an order- N tensor as a sum of R rank-one terms [Kolda and Bader, 2009]:

$$\mathcal{X} \approx \sum_{r=1}^R \mathbf{a}_r^{(1)} \circ \mathbf{a}_r^{(2)} \circ \dots \circ \mathbf{a}_r^{(N)}. \quad (4.1)$$

For the V1 subject-channel-frequency-time tensor, the subject factor can be used as a compact embedding. Non-negativity improves parts-based interpretability for non-negative power data, but it is an optimization constraint, not a guarantee of identifiability or classification performance.

The preliminary graph-regularized V1 objective has the conceptual form

$$\min_{A, B, C, D \geq 0} \frac{1}{2} \|\mathcal{X} - [[A, B, C, D]]\|_F^2 + \lambda \text{tr}(B^T L B), \quad (4.2)$$

where B is the channel factor and L is an electrode-graph Laplacian. The implemented multiplicative update mixes reconstruction and graph penalties, but its recorded objective scales those terms inconsistently under factor renormalization. The corresponding result is retained as preliminary rather than presented as validated graph regularization.

4.3 Tucker decomposition and HOSVD

Tucker decomposition uses a core tensor and one factor matrix per mode:

$$\mathcal{X} \approx \mathcal{G} \times_1 U^{(1)} \times_2 U^{(2)} \dots \times_N U^{(N)}. \quad (4.3)$$

Unlike CP's single component count, Tucker can choose a different multilinear rank for each mode. V2 implements a training-only HOSVD-like projection: each factor is formed from leading eigenvectors of a mode covariance, and their Kronecker-structured combination defines a reusable orthonormal basis. V3 uses channel and lag factors to form waveform-energy filters. Tucker is a natural course bridge from matrix PCA to structured higher-order compression [Kolda and Bader, 2009, Padhy, 2026b].

4.4 Tensor Train decomposition

Tensor Train writes each entry as a product of small cores:

$$x_{i_1 \dots i_N} = G_1[i_1]G_2[i_2] \cdots G_N[i_N], \quad (4.4)$$

with boundary ranks equal to one. TT-SVD obtains cores from successive low-rank unfoldings and controls storage through intermediate TT ranks [Oseledets, 2011]. In V2, feature modes are placed before observations; left-orthogonal cores define a common training-derived basis. This avoids comparing separately factorized subjects whose cores may differ by arbitrary signs, rotations, or gauges.

For the V2 $19 \times 59 \times 3$ window tensor, the basis is assembled as

$$B_{(i,j,k),d} = \sum_{a,b} G_1(1, i, a)G_2(a, j, b)G_3(b, k, d). \quad (4.5)$$

A centered held-out tensor is projected by matrix multiplication with B ; the basis is never refit on held-out people.

4.5 Welch and spectrogram tensorization

Welch’s method averages modified periodograms from windowed, overlapping segments to reduce spectral-estimate variance [Welch, 1967]. The project uses a 500-sample Hann segment at 250 Hz, 250-sample overlap, and 500-point FFT, giving 0.5 Hz bins. Frequencies from 1 to 30 Hz yield 59 bins.

The V1 relative-power map for subject n , channel c , frequency f , and time bin q can be written as

$$X_{ncfq} = \text{median}_{w \in q} \sqrt{\frac{P_{ncfw}}{\sum_{c',f'} P_{nc'f'w} + \epsilon}}. \quad (4.6)$$

V2 instead preserves signed log absolute power:

$$Z_{w,c,f,s} = \log_{10} \left(\max(P_{w,c,f,s}, 10^{-12}) \right), \quad (4.7)$$

where $s \in \{1, 2, 3\}$ indexes the three local overlapping two-second segments within a four-second window. This representation can express increases and decreases after centering, so non-negativity is neither required nor desirable.

4.6 Waveform-delay tensor and exact moment compression

Each clean four-second window is anti-alias resampled from 250 to 125 Hz, producing 500 samples. A 16-lag sliding view forms $\mathcal{D} \in \mathbb{R}^{19 \times 16 \times 485}$, where lag spacing is 8 ms and the first-to-last span is 120 ms. Each channel-lag row is centered over local positions.

Let $z_p \in \mathbb{R}^{304}$ be the vectorized channel-lag sample at local position p . The per-window moment is

$$S_w = \frac{1}{485} \sum_{p=1}^{485} z_p z_p^T, \quad (4.8)$$

and the subject moment is the equal-window average. For any filter $b \in \mathbb{R}^{304}$, its mean squared response is exactly

$$E(b) = \frac{b^T S b}{b^T b}. \quad (4.9)$$

Thus the stored $88 \times 304 \times 304$ moments are exact sufficient statistics for the pooled squared response of any linear delay filter. They avoid storing all delay tensors, but they deliberately discard local-position and long-range sequence order. The waveform branch is therefore a pooled second-order model, not a general raw-EEG neural sequence model.

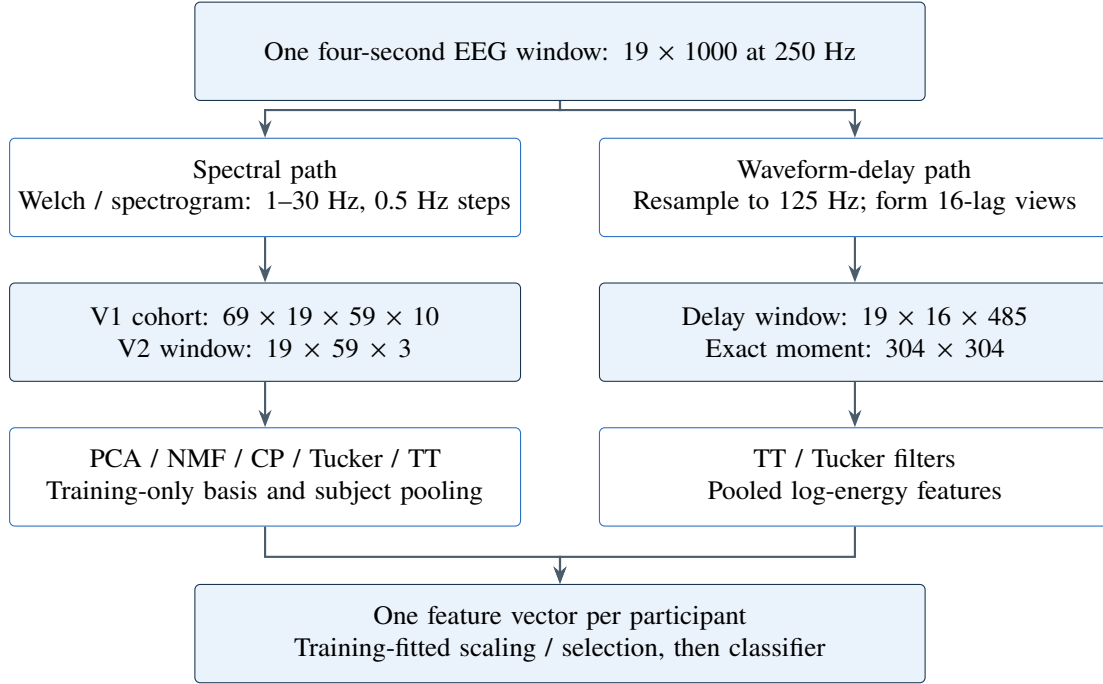


Figure 4.1: Implemented spectral and waveform-delay tensorizations, with their actual array dimensions. The two branches are alternative representations; fusion is evaluated separately in later experiments.

5 System Architecture and Implementation Design

5.1 End-to-end component architecture

The repository uses versioned, append-only stages. Source BIDS and EEGLAB files feed deterministic preprocessing partitions; tensor preparation creates new arrays and indexes; each benchmark freezes its protocol, persists fold-level models and predictions, and runs a separate verification/report step. Figure 5.1 shows the implementation-level data flow.

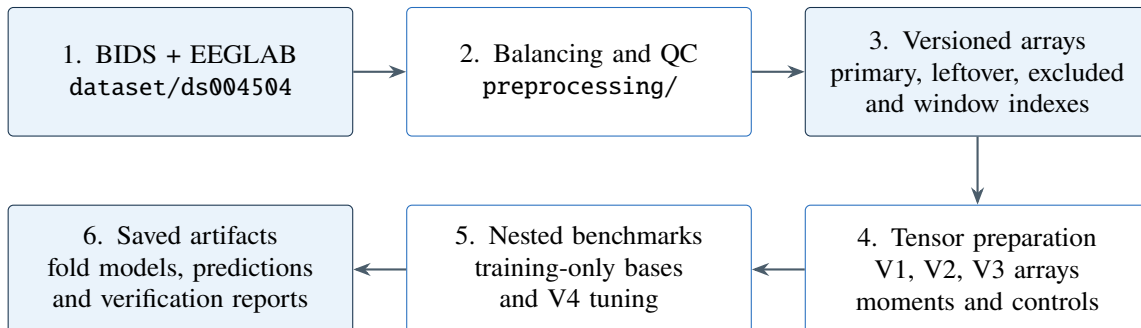


Figure 5.1: Implementation architecture. Follow steps 1–6 across the top row, then back along the bottom row. Derived stages use versioned directories; source files remain read-only.

5.2 Core preprocessing components

preprocessing/build_balanced_dataset.py

Loads participant and recording metadata, down-samples derivatives, scans complete windows, constructs FTD duration references, solves the SciPy MILP assignment, spreads chosen windows in time, writes signals/metadata/manifests, preserves leftovers and tails, and verifies the partition.

preprocessing/audit_primary_signal_quality.py

Computes robust per-window, per-channel, and per-subject signal statistics, spectral band summaries, shared-variance diagnostics, and candidate outlier flags without modifying the primary arrays.

preprocessing/build_clean_primary_dataset.py

Applies CAR and mean removal, assigns post-CAR exclusion reasons, performs deterministic medium-bucket balance trims, writes retained and excluded partitions, and verifies finiteness, group equality, means, and source partition completeness.

5.3 Tensor and model components

tensor_pipeline/tensorize_subjects.py

Builds the V1 non-negative relative-power tensor with 10 chronological rank bins and saves channel, frequency, subject, and label arrays.

tensor_pipeline/run_decomposition_benchmark.py

Implements PCA, NMF, non-negative CP, graph NCP, and non-negative Tucker, including held-out non-negative least-squares projection. The CPU launcher swaps the optional XGBoost classifier for histogram gradient boosting while retaining decomposition behavior.

tensor_pipeline/v2/prepare.pyandmodels.py

Corrects electrode order, enforces boundary guards, constructs signed logPSD tensors and 418 controls, and implements reusable TT/HOSVD projections, anatomical smoothing, classifier grids, and trainable pipelines.

tensor_pipeline/v2/benchmark.py

Runs nested participant CV, fits decomposition bases using training people, selects configurations from inner out-of-fold predictions, persists bases and classifiers, and aggregates complete repeats.

tensor_pipeline/v3_waveform/prepare.py

Together with `wave_models.py`, this component rebuilds the full-cohort analysis index from primary plus leftovers, computes exact waveform-delay moments and same-cohort spectral controls, and implements unsupervised TT/Tucker energy projections, subject weights, and supervised TT.

tensor_pipeline/v4_tuning/tune_models.py

Creates the V4 feature bank, stronger-regularized TT objective, 19 classifier settings, seven representation settings, and six supervised TT settings.

tensor_pipeline/v4_tuning/benchmark_tuning.py

Freezes and evaluates 139 configurations in nested CV, produces automatic and per-family selections, stores training-only bases and full pipelines, and records every held-out prediction.

5.4 Artifact schema

Each major stage writes both numerical arrays and human/machine-readable provenance:

- `signals.npy` or feature arrays for efficient memory mapping;
- `metadata.npz` or `index.npz` for subject IDs, labels, sample IDs, window positions, counts, and channel names;

- JSON protocol, preparation, selection, verification, and fingerprint documents;
- fold-level JSON predictions and inner scores;
- trusted-local joblib models and bases; and
- portable XGBoost booster exports accompanied by their required complete preprocessing pipelines.

5.5 Leakage barriers by construction

The most important architectural boundary is participant membership. A transformation T is learned as $T_{T_{\text{train}}}$ from training participants only, then applied unchanged to validation or test participants. This rule covers decomposition means and bases, graph-pre-smoothed bases, univariate feature selection, standardization, class weights, classifier fitting, and supervised TT parameters.

In V2, exactly 24 windows per training subject are spread across that subject to calibrate window-level decomposition bases. All of a held-out person’s windows are projected afterward. In V3/V4, per-subject delay moments and spectral controls are fixed input summaries, while decomposition, feature selection, and scaling are fit inside the relevant training fold.

5.6 Computational design choices

The completed runs are CPU-oriented. Memory maps prevent loading all waveform windows simultaneously. Covariance eigensolves replace explicit huge unfolding SVDs where mathematically equivalent. Delay moments compress waveform-filter energy evaluation from all $19 \times 16 \times 485$ tensors to 304×304 matrices per subject. Thread pools are limited to four workers to avoid nested parallel oversubscription. V4 uses XGBoost’s CPU histogram tree method; the installed build reports `USE_CUDA=false`. PyTorch and the GPU were not required for the completed experiments.

6 Experiment 1: Exploratory Spectral Tensor Benchmark

6.1 Objective and input cohort

The first experiment tests whether a compact non-negative subject-channel-frequency-time tensor can support three-class prediction on the controlled 69-person cohort. It uses the 12,270-window clean primary partition before the later EEGLAB boundary correction. The experiment is valuable as a baseline and design diagnostic, but its maximum across ranks and models is exploratory because the outer score is also used to compare those choices.

6.2 Tensor construction

For every four-second window and each of 19 channels, Welch PSD uses a 500-sample Hann segment, 250-sample overlap, 500-point FFT, constant detrending, and density scaling. The 59 bins from 1 to 30 Hz are retained. Power is normalized over the whole channel-frequency map and square-root transformed to obtain a non-negative representation.

Each participant’s windows are sorted chronologically and assigned to 10 rank-based time bins. The median spectral map within each bin produces

$$\mathcal{X} \in \mathbb{R}_+^{69 \times 19 \times 59 \times 10}. \quad (6.1)$$

The 10 bins summarize early-to-late recording position; they are not event- aligned epochs across people.

6.3 Decomposition methods

The benchmark evaluates ranks 3, 5, and 8 for five representations:

1. PCA on the flattened tensor, fitted on training participants;
2. NMF with NNDSVDa initialization, coordinate descent, 1,000 iterations, tolerance 10^{-5} , and held-out transform;
3. non-negative CP fitted by multiplicative updates;
4. graph-regularized non-negative CP with nominal graph weight 0.1; and
5. non-negative Tucker with subject rank equal to the experimental rank and feature ranks capped at channel 8, frequency 12, and time 5.

CP/Tucker held-out features are solved in a fixed training-derived basis using non-negative least squares. Decomposition fitting occurs separately inside each subject-level training fold.

6.4 Classifiers and validation

Each representation is paired with regularized multinomial logistic regression, RBF SVM, and histogram gradient boosting. Logistic and SVM use standardized features and balanced class weights. The evaluation uses three repetitions of stratified five-fold participant CV with seed 5040. Since each tensor row is one participant, there is no window-level split at this stage.

6.5 Results

Representation	Best rank	Mean BA	Associated classifier
PCA	8	54.1%	Logistic regression
NMF	8	54.1%	Logistic regression
Non-negative Tucker	3	53.6%	Histogram gradient boosting
Non-negative CP	5/8	52.2%	Logistic regression
Graph non-negative CP	5	51.7%	Logistic regression

Table 6.1: Strongest observed V1 result for each representation family. These are exploratory maxima, not nested-selection estimates.

For the cited PCA rank-8 result, mean BA is 54.1% with repeat SD 3.6 percentage points. Its probability-averaged 69-subject confusion matrix has recalls of 52.2% AD, 30.4% FTD, and 73.9% CN. The low FTD recall immediately shows that equal window counts do not eliminate the difficulty of the smallest diagnostic group.

An NMF convergence check raises the rank-8 iteration limit to 5,000 and produces the same classification result. Reconstruction error improves with rank for CP, but prediction does not improve monotonically. This separates compression fidelity from discriminative utility.

6.6 Feature-bottleneck diagnostic

A later diagnostic constructs less-compressed channel-band-time features and uses repeated five-fold CV with training-only univariate selection. Its strongest combination, 760 raw features reduced to 40 and classified by a linear SVM, reaches 57.0% mean balanced accuracy. This suggests that aggressive low-rank compression may remove some useful structure, but it is not a causal ablation because feature form and selection also change.

6.7 Known limitations and correction path

The first tensorizer attached a hard-coded electrode name order to rows that were actually stored in another EEGLAB order. Consequently, the graph edges and any electrode-specific interpretation in V1 are invalid. PCA, flattened NMF, global statistics, and channel-permutation-invariant accuracy calculations are not invalidated by a consistent row permutation. The graph-NCP objective also has a factor-scale inconsistency. V2 therefore reads headers directly, uses signed absolute logPSD, guards boundaries, and evaluates a fully nested pipeline.

7 Experiment 2: Corrected Nested Tensor Train and Tucker Benchmark

7.1 Corrections and representation changes

V2 is the first fully corrected tensor benchmark. It verifies the 19-channel EEGLAB row order for all 69 controlled-cohort participants, applies the 0.5 s boundary guard, and retains 11,942 windows. It also replaces relative, non-negative power with signed log absolute PSD so that centered increases and decreases remain available to an orthogonal low-rank model.

Each retained window is represented as $\mathcal{Z}_w \in \mathbb{R}^{19 \times 59 \times 3}$: channel $\times$ frequency $\times$ local segment. A separate subject tensor stores mean, median, and standard deviation logPSD, also of shape $19 \times 59 \times 3$. The local segment axis is not described as shared event time.

7.2 Spectral controls

The V2 control vector has 418 features. For delta (1–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), and beta (13–30 Hz), it records per-channel mean and standard deviation of log absolute band power and relative band power. It also records per-channel mean and standard deviation of spectral entropy, 7–13 Hz alpha peak, and log slow-to-fast power ratio. Age, sex, MMSE, duration, window count, and diagnosis-derived statistics are absent.

7.3 Inductive TT and Tucker projections

The `TTProjection.fit()` implementation centers its training tensor, moves observations to the last mode, and computes leading left vectors from AA^T for each unfolding. Signs are fixed by the largest-magnitude element for reproducible cores. The contracted cores form a common orthonormal basis; the code records orthogonality and centered reconstruction error.

Window-level bases use 24 chronologically spread calibration windows per training participant. For every participant, all projected windows are pooled using four statistics per latent component: mean, standard deviation, median, and interquartile range. The decomposition basis is trained only from the relevant training participants; calibration provenance is saved.

Tucker uses training-only multilinear PCA/HOSVD ranks (6, 8, 3). Graph TT first applies a fixed coordinate-derived smoother. Each electrode connects to its four nearest coordinate neighbors with Gaussian distance weights. For normalized Laplacian $L = I - D^{-1/2}AD^{-1/2}$ and strength $\eta = 0.5$, the smoother is

$$S = V \operatorname{diag} \left((1 + \eta \lambda_i)^{-1/2} \right) V^T, \quad (7.1)$$

where $L = V \operatorname{diag}(\lambda_i) V^T$. This is deterministic anatomical pre-smoothing, not the preliminary graph penalty in Eq. 4.2.

7.4 Candidate families and models

Six representation families are compared:

1. subject-level TT at ranks 8, 16, and 32;
2. pooled window TT at ranks 8, 16, and 32;
3. graph-smoothed pooled window TT at ranks 8, 16, and 32;
4. pooled window Tucker with training-only selection of 16, 32, or 64 latent features;
5. TT plus 418 spectral controls with TT ranks 8, 16, or 32 and selection of 32 combined features; and
6. spectral controls with selection of 16, 32, or 64 features.

Classifier candidates are logistic regression with $C \in \{0.1, 1, 10\}$, linear SVM with $C \in \{0.1, 1\}$, RBF SVM with $C \in \{1, 10\}$, and a fixed shallow histogram gradient booster. Feature selection uses ANOVA F scores; every selector and scaler is inside the fitted pipeline.

7.5 Nested subject cross-validation

The outer evaluation is three repetitions of stratified five-fold participant CV (seed 5040). Each outer training set contains an independent three-fold inner CV (seed 8000 plus outer index). Inner out-of-fold balanced accuracy, then macro F1, selects rank, representation, feature count, and classifier. Exact ties use a deterministic lower-complexity ordering. This design addresses the optimistic bias produced when the same cross-validation result both selects and evaluates a model [Varma and Simon, 2006].

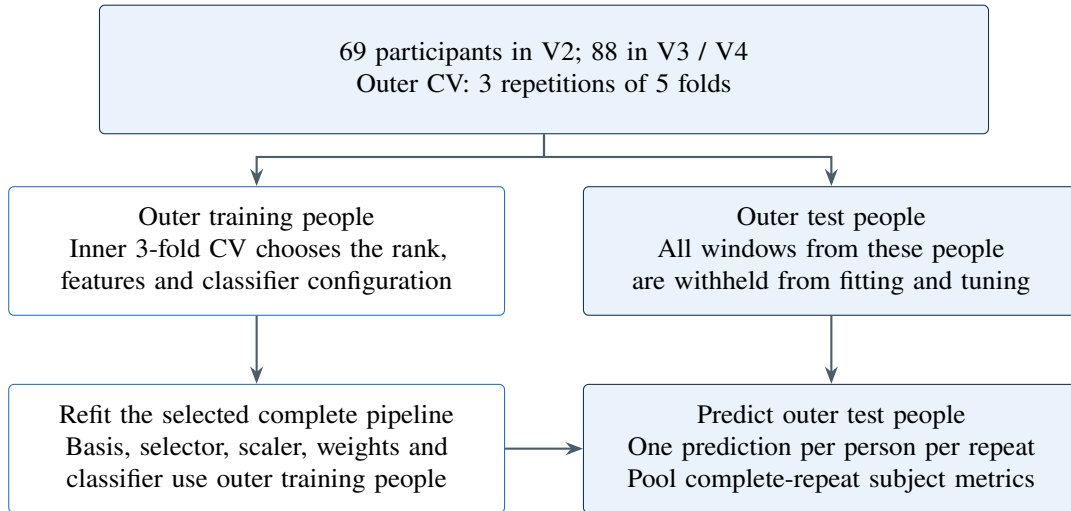


Figure 7.1: Nested participant-level training, validation and test preparation in V2–V4. Inner validation selects a configuration; outer test labels enter only the final scoring step.

The *primary* V2 endpoint is the outer score of the entire inner-selected pipeline. Per-family outer results are secondary method comparisons. SVM class labels come from `predict()`, while its decision scores are stored as uncalibrated decision functions rather than probabilities.

7.6 Results and interpretation

The strongest observed family is Tucker/HOSVD at 56.0% mean balanced accuracy, with mean recalls of 59.4% AD, 50.7% FTD, and 58.0% CN. However, the complete automatic-selection procedure reaches only 47.3%. The gap shows selection variance in a small cohort: selecting among many near-tied pipelines inside roughly 55 training participants can fail to identify the family that looks strongest after aggregation.

7.7 Verification

Numerical tests compare TT reconstruction against TensorLy, verify orthogonality, recover full-rank synthetic tensors, and check batch-invariant transform output. Post-run verification replays 105 saved

V2 family	Mean BA	Repeat SD
Inner-selected complete pipeline (primary)	47.3%	–
Window Tucker / HOSVD	56.0%	3.3 pp
TT plus spectral controls	54.1%	–
Graph-smoothed window TT	53.1%	–
Spectral controls	51.7%	–
Window TT	51.2%	–
Subject TT	50.2%	3.0 pp

Table 7.1: Corrected V2 nested subject-level results. Dashes indicate values not emphasized in the consolidated record.

classifier pipelines, 60 decomposition sets, 45 inner splits, and 15 outer splits. Held-out features are recomputed from saved training bases and must reproduce predictions and scores. Input hashes and source transforms are checked; no test participant enters basis calibration.

8 Experiment 3: Full-Cohort Waveform Delay-Tensor Analysis

8.1 Why use the leftovers?

The controlled cohort is valuable for comparability but excludes signal from 19 additional people and thousands of valid windows. V3 therefore creates a new analysis index over both stage-02 primary and leftover partitions. It does not merge or overwrite those partitions. Uniform CAR, QC, and boundary rules yield 16,824 retained windows from all 88 people.

The key imbalance rule changes from equal window counts to equal *person* influence. For training label y_i , the subject weight is

$$\omega_i = \frac{N_{\text{train}}}{3N_{\text{train}, y_i}}. \quad (8.1)$$

Each diagnosis receives one third of total loss mass. Within a person, windows are averaged, not summed; equivalently, each window contributes its person’s weight divided by that person’s retained-window count. There is no duplication, SMOTE, or synthetic FTD data.

8.2 Waveform branch

The waveform representation is the delay tensor and exact moment method from Eq. 4.9. The unsupervised projector first forms a class- balanced training moment. For TT, a channel factor of rank 6 is obtained from the channel marginal; a second eigensolve in the reduced channel-lag space produces 8 or 16 filters. Tucker uses channel and lag ranks (4, 4) or (6, 6). Log pooled filter energies become features.

The code numerically checks that moment-derived energy equals explicit raw delay-tensor contraction and that its TT factors agree with an equivalent TT-SVD construction on synthetic data.

8.3 Supervised TT

The supervised model parameterizes filters as

$$b_{c\ell k} = \sum_{r=1}^{R_c} U_{cr} V_{r\ell k}, \quad (8.2)$$

then maps standardized log energies to three softmax logits. V3 jointly optimizes U , V , classifier head H , and bias using class-weighted cross-entropy:

$$\mathcal{L} = \sum_i \tilde{\omega}_i \left[\log \sum_{g=1}^3 e^{q_{ig}} - q_{i,y_i} \right] + \frac{\lambda_H}{2} \|H\|_F^2 \quad (8.3)$$

$$+ \frac{10^{-3}}{2} \left(\|U - U_0\|_F^2 + \|V - V_0\|_F^2 \right). \quad (8.4)$$

The training-only unsupervised TT initializes U_0, V_0 and energy scaling. L-BFGS-B has a fixed 100-iteration budget; head penalties 0.01 and 0.1 are inner CV candidates. A reported convergence flag is an optimizer status, not a proof of global optimality for this nonconvex problem.

8.4 Spectral control and candidate set

The same 418 spectral definitions as V2 are recomputed on all retained V3 windows. Training-only ANOVA selects 16 or 32. Wave TT ranks 8/16, waveform Tucker ranks 4/6, and the spectral settings are each paired with logistic regression ($C = 0.1, 1$) or RBF SVM ($C = 1, 10$). Together with two supervised TT penalties, V3 has 26 candidate configurations.

Evaluation uses nested three-repeat five-fold outer / three-fold inner participant CV, outer seed 9060 and inner seed 90600+ outer index. The primary endpoint is again the complete inner-selected pipeline.

8.5 Results

V3 family	Mean BA	Accuracy	Macro F1
Inner-selected complete pipeline (primary)	52.0%	—	—
Spectral control	55.8%	—	—
Waveform Tucker	51.8%	—	—
Waveform TT	50.9%	—	—
Supervised TT	49.9%	—	—

Table 8.1: Full-cohort V3 results. The spectral control is the strongest observed family.

The negative result is informative: more raw-looking waveform structure does not automatically improve classification. The delay representation captures short-lag second-order energy but removes absolute phase, local ordering, and long-range temporal dynamics. The spectral control may better match the stable disease-related signal available in this small cohort.

8.6 Optimization-budget sensitivity

All 120 supervised TT fits in the original V3 benchmark reach the 100-iteration limit. A post-hoc sensitivity keeps the already specified head penalty 0.1 and compares 100 with 2,000 iterations on the same 15 full outer-training splits and 10 smaller learning-curve subsets. Across full splits, 100 iterations averages 55.0% balanced accuracy and 2,000 averages 53.7%. Only 3 of 25 long-budget fits report convergence. More iterations alone therefore do not resolve the model’s weakness; these values are diagnostic and do not replace the V3 primary endpoint.

The fixed-model learning curve uses approximately 27, 48, and 70 training people and obtains 42.0%, 40.9%, and 55.2% held-out balanced accuracy. The curve is non-monotone and based on one CV repetition, so it cannot estimate the number of additional participants required.

8.7 Verification

V3 checks reconstruct sampled source windows, confirms channel order and class weight totals, replays saved models and scores, verifies subject-disjoint splits, preserves every V2 window, excludes all prior artifact IDs, and fingerprints the unchanged primary and leftover arrays.

9 Experiment 4: Bounded Classifier and Optimization Tuning

9.1 Frozen scope

V4 uses exactly the V3 88-person inputs, delay moments, spectral controls, and outer subject splits. No new EEG exclusion or transformation is introduced. Before new outer outcomes are examined, 139 configurations are saved in `protocol.json`. This bounds researcher degrees of freedom and makes the reported tuning reproducible.

9.2 Representation grid

Seven unsupervised feature settings are crossed with 19 classifiers:

- waveform TT terminal rank 8 or 16, channel rank 6;
- waveform Tucker channel/lag rank 4 or 6;
- 418 spectral controls with training-only selection of 16 or 32;
- TT16 concatenated with 418 spectral controls, followed by training-only selection of 32 combined features.

The fusion setting does not guarantee use of TT features. ANOVA selection can choose mostly or entirely spectral variables. A fusion win is therefore evidence for the complete representation-selection pipeline, not isolated proof of a tensor contribution.

9.3 Classifier grid

The 19 settings are:

- logistic regression with $C \in \{0.01, 0.1, 1\}$;
- RBF SVM with $C \in \{0.1, 1, 10\}$ and $\gamma \in \{\text{scale}, 0.01\}$ [Cortes and Vapnik, 1995];
- eight fixed XGBoost combinations spanning depth 1–3, learning rate 0.03–0.1, 100–250 trees, minimum child weight 1–5, L_2 penalty 1–10, L_1 penalty 0–0.5, row subsampling 0.8–1.0, and column subsampling 0.8 [Chen and Guestrin, 2016]; and
- two 200-tree random forests: depth 3/minimum leaf 3, or unrestricted depth/minimum leaf 5, with square-root feature sampling [Breiman, 2001].

All pipelines standardize features, including tree models, for one consistent persisted schema. Classifier fitting receives the weights in Eq. 8.1. XGBoost uses CPU histogram trees, multiclass soft probabilities, fixed tree counts, and no test-driven early stopping.

9.4 Stronger-regularized supervised TT

Six additional supervised TT settings use channel rank 2 or 4, 4 or 8 filters, head penalty 0.1 or 1, core anchor 0.1 or 1, and a fixed 500-iteration budget. The altered objective replaces the weak 10^{-3} anchor from V3 with a much stronger constraint toward the training-only unsupervised TT initialization. Analytic gradients are numerically checked before benchmarking.

9.5 Selection hierarchy

For each outer fold, three-fold inner out-of-fold predictions rank all 139 configurations by balanced accuracy, then macro F1, then deterministic lower complexity. The code saves ten outer entries:

- one complete inner-selected pipeline (the primary endpoint);
- best inner-selected member of each of five classifier families; and
- best inner-selected member of each of four representation families.

These are not ten statistically independent fitted models; several named views can reference the same fitted configuration. Per-family scores are secondary comparisons and cannot replace the primary result after test inspection.

9.6 Software execution

The official `xgboost-cpu==3.2.0` package was installed into the existing Python 3.10 user environment without upgrading dependencies, creating a virtual environment, installing PyTorch, or enabling CUDA. The fitted XGBoost build reports OpenMP enabled and CUDA disabled. Nested fitting took 316.3 seconds, excluding installation and verification. Core estimators use scikit-learn’s pipeline interfaces [Pedregosa et al., 2011].

9.7 Results

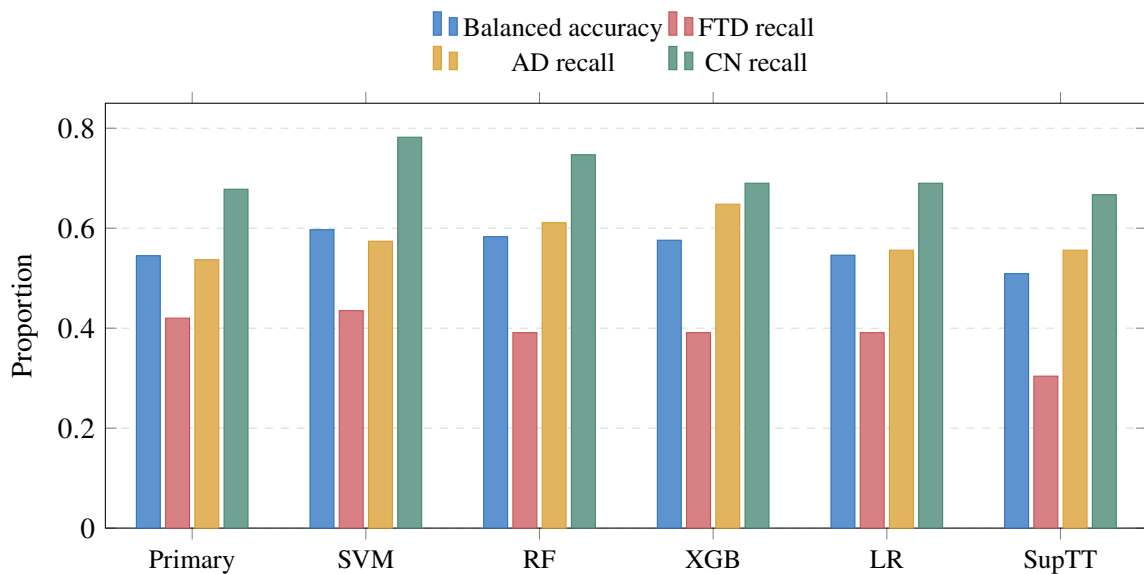


Figure 9.1: V4 classifier-family balanced accuracy and diagnosis recall. SVM is the strongest observed family, while FTD recall remains the weakest class-specific measure.

The complete automatic-selection endpoint is 54.5% mean balanced accuracy (repeat SD 3.2 pp), ordinary accuracy 55.3%, and macro F1 0.540. It improves nominally by 2.5 percentage points over V3’s 52.0% primary endpoint on the same inputs and outer splits, but adaptive reuse of the cohort prevents treating this as independent confirmation.

The strongest observed classifier family is RBF SVM at 59.7% mean balanced accuracy (repeat SD 0.9 pp), ordinary accuracy 60.6%, and macro F1 0.589. Random forest reaches 58.3%; XGBoost 57.6%; logistic regression 54.6%; and stronger-regularized supervised TT 50.9%. SVM is stronger than XGBoost in this bounded comparison.

Representation-family results are TT/spectral fusion 59.1% (SD 3.6 pp), spectral 56.8% (SD 1.2 pp), waveform Tucker 52.0% (SD 2.3 pp), and waveform TT 46.8% (SD 11.7 pp). The large TT repeat variability is a warning against selecting a favorable repeat or fold.

9.8 Optimization and verification

Among 285 supervised optimization fits, 200 report convergence and 85 reach the 500-iteration limit. This is an optimization-status improvement over V3 but not a predictive win. Verification passes all of the following:

- 150 saved outer model entries replayed with matching scores;
- 60 training-only basis sets and 45 inner plus 15 outer splits checked;
- exact inner selections, scaler means, and class-weight totals reproduced;
- 23 XGBoost boosters exported and reloaded with matching outputs; and
- original primary and leftover array hashes unchanged.

10 Integrated Results and Error Analysis

10.1 Cross-experiment view

Table 10.1 collects the principal and strongest observed results without treating them as directly exchangeable. V1 and V2 use the controlled 69-person cohort; V2 adds boundary guards. V3 and V4 use all 88 people and a different fold seed. V3 and V4 are the cleanest paired comparison because they share the same input arrays and outer splits.

Stage	Reported procedure	Mean BA	Status
V1	PCA/NMF rank 8 plus logistic regression	54.1%	Exploratory maximum
V2	Complete inner-selected pipeline	47.3%	Primary endpoint
V2	Tucker/HOSVD family	56.0%	Secondary family
V3	Complete inner-selected pipeline	52.0%	Primary endpoint
V3	Same-cohort spectral control	55.8%	Secondary family
V4	Complete inner-selected pipeline	54.5%	Primary endpoint
V4	Tuned RBF SVM family	59.7%	Secondary family
V4	TT/spectral fusion family	59.1%	Secondary family

Table 10.1: Cross-experiment results and their correct inferential status.

The 59.7% SVM value is the highest retained result, but it is not an accuracy ceiling and cannot be promoted over the V4 primary endpoint. The project has no untouched external test cohort. Repeated outer folds also share much of their training data, so the repeat SD is descriptive rather than a confidence interval.

10.2 Metric definitions

For confusion matrix M with classes $g \in \{\text{AD}, \text{FTD}, \text{CN}\}$, class recall is

$$R_g = \frac{M_{gg}}{\sum_j M_{gj}}, \quad (10.1)$$

and balanced accuracy is

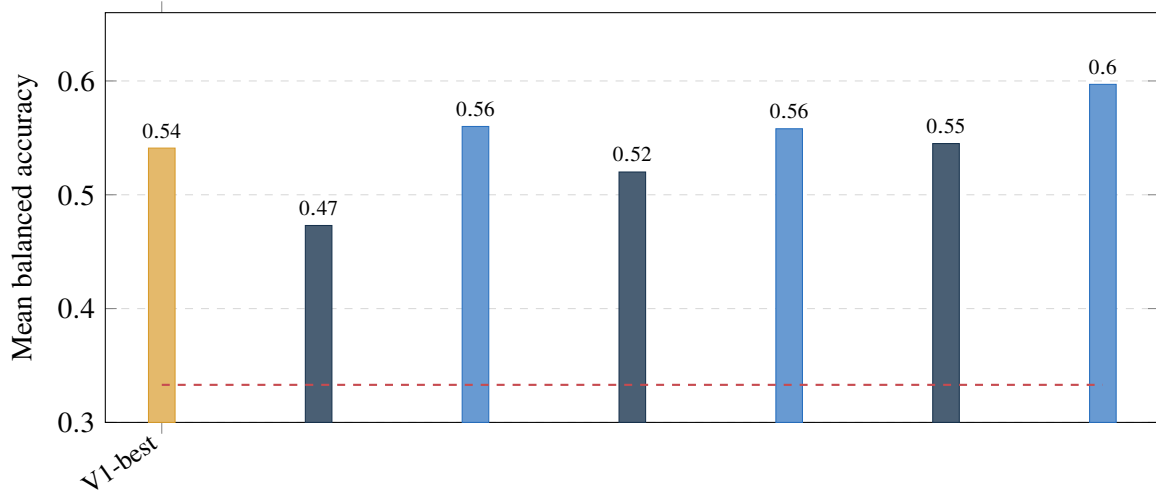


Figure 10.1: Experiment progression. Navy bars are predeclared complete-pipeline endpoints; blue bars are secondary best-family results; the V1 maximum is exploratory. The dashed baseline is 33.3% balanced accuracy.

$$BA = \frac{1}{3} \sum_g R_g. \quad (10.2)$$

Always predicting one class gives 33.3% balanced accuracy. In the 88-person cohort, always predicting AD gives 40.9% ordinary accuracy because AD is the largest class. Macro F1 gives each class equal weight after combining precision and recall, making it useful alongside BA.

10.3 Diagnosis-specific error pattern

FTD is the consistent weakness. In the V1 PCA result, FTD recall is 30.4%. V2 Tucker improves it to 50.7%, but V4 classifier-family values remain 30.4–43.5%. The strongest V4 SVM family has mean recalls of 57.4% AD, 43.5% FTD, and 78.2% CN. The model is therefore much better at recognizing controls than at separating the two dementia groups.

This pattern is plausible for three non-exclusive reasons:

1. **Independent sample limitation.** Only 23 FTD people exist. Thousands of additional four-second windows stabilize each person’s features but do not provide new independent anatomy or disease variability.
2. **Phenotypic overlap.** Resting spectral slowing and covariance changes need not uniquely separate AD from FTD in a small heterogeneous routine-EEG cohort.
3. **Representation mismatch.** The strongest features may emphasize healthy-versus-dementia differences, while the finer AD-versus-FTD distinction may require connectivity, topography, longer temporal structure, or external clinical information not used here.

10.4 What tensorization contributed

The results support a nuanced conclusion:

- Tensorization is useful for imposing explicit channel-frequency-time or channel-lag structure and for testing low-rank hypotheses.
- Tucker/HOSVD is the strongest corrected V2 family at 56.0%, showing that multilinear compression can be competitive.

- V3 spectral controls outperform every waveform-only branch, so a more “raw” representation is not automatically better.
- V4 fusion reaches 59.1%, but its feature selector can choose spectral dimensions; the experiment does not prove an isolated TT gain.
- Waveform TT has high instability in V4 (11.7 pp repeat SD), suggesting sensitivity to the available training people and selected classifier.

The correct claim is that structured tensor methods were implemented, tested, and compared under leakage-aware validation. The data do not support a claim that tensor decomposition consistently dominates spectral baselines.

10.5 Classifier effects

V4 shows that classifier choice matters within the existing representations. RBF SVM is strongest at 59.7%, with a small descriptive repeat SD. Random forest and XGBoost follow at 58.3% and 57.6%. The nonlinear kernel appears better matched to the selected low-dimensional feature geometry than the tested boosted trees. XGBoost remains useful because it rules out the assumption that a modern tree ensemble must be best on this small participant-level table.

The automatic pipeline selects SVM in 7 outer fits, logistic regression in 5, and random forest, XGBoost, and supervised TT once each. It selects fusion in 7 folds, spectral features in 7, and supervised TT once. This heterogeneity explains why a single globally attractive family may still differ from the complete fold-wise primary endpoint.

10.6 Optimization effects

V3’s supervised TT reaches its 100-iteration cap in every original fit. Raising the cap to 2,000 does not improve the fixed-model mean. V4 instead changes the objective by strengthening the core anchor and reducing ranks, leading to 200/285 reported convergences. Predictive performance remains only 50.9% for the supervised TT family. Therefore:

Better numerical convergence is neither sufficient nor synonymous with better generalization. Optimization quality, model bias, and data information content must be evaluated separately.

10.7 Was more data needed?

Using leftovers is worthwhile because it adds 19 people and 4,828 valid leftover windows to the full-cohort analysis. Yet the main shortage remains independent FTD participants. Future data collection should prioritize new FTD people from an external site, plus an untouched validation cohort. Extra windows from the same 23 people have diminishing value after per-person features stabilize.

The existing learning curve is too small and noisy to specify a required sample size. A defensible sample-size study would use learning curves repeated over many participant subsamples, preserve the three-class endpoint, and quantify uncertainty. Until then, a statement such as “we need exactly 50 more people” would be invented rather than evidence-based.

10.8 Fairness and confounding checks

The controlled cohort broadly matches age, sex, duration buckets, people, and window counts while retaining natural variation. The full cohort uses all available people and corrects only loss weighting. Neither design uses protected or clinical metadata as predictors. Nevertheless, the dataset is a single-site clinical cohort and subgroup performance by age/sex has not been externally validated. Demographic audit reduces obvious confounding; it does not establish fairness or generalizability.

11 Reproducibility, Verification, and Software Environment

11.1 Reproducibility strategy

Reproducibility is implemented through versioned directories, frozen protocols, deterministic seeds, source fingerprints, explicit sample IDs, saved split memberships, persisted bases and complete classifier pipelines, fold-level JSON predictions, and separate replay verification. Earlier experiments are retained even when corrected later.

11.2 Recorded software environment

The V2 artifact manifest records Python 3.10.11, NumPy 2.2.6, SciPy 1.15.3, scikit-learn 1.7.2, Joblib 1.6.0, TensorLy 0.9.0, and threadpoolctl 3.6.0. V4 adds `xgboost-cpu` 3.2.0. MiKTeX 25.12 is used only to compile this report; it is not part of the modeling environment.

11.3 Deterministic settings

Key seeds are:

Purpose	Seed	Stage
First primary shuffle	4504	Stage 02
First leftover shuffle	4505	Stage 02
Clean primary shuffle	4506	Stage 04
Clean excluded shuffle	4507	Stage 04
V1 repeated subject CV	5040	Experiment 1
V2 outer subject CV	5040	Experiment 2
V2 inner subject CV	8000 + outer index	Experiment 2
V3/V4 outer subject CV	9060	Experiments 3–4
V3/V4 inner subject CV	90600 + outer index	Experiments 3–4
Tree models	9064	Experiment 4

Table 11.1: Principal deterministic seeds.

Determinism does not eliminate uncertainty. These seeds make a specific analysis replayable; they do not guarantee that another reasonable split seed gives the same score.

11.4 Recommended execution order

The versioned scripts should be run in dependency order from the workspace root:

1. `py -3.10 preprocessing/build_balanced_dataset.py`
2. `py -3.10 preprocessing/audit_primary_signal_quality.py`
3. `py -3.10 preprocessing/build_clean_primary_dataset.py`
4. `py -3.10 tensor_pipeline/tensorize_subjects.py` and the V1 rank benchmarks if historical reproduction is required;
5. `py -3.10 tensor_pipeline/v2/prepare.py`, then V2 tests, benchmark, verification, and reporting;
6. `py -3.10 tensor_pipeline/v3_waveform/prepare.py`, then V3 tests, run, convergence sensitivity, verification, and finalization; and

7. `py -3.10 tensor_pipeline/v4_tuning/test_tuning.py`, then V4 benchmark and verification/report scripts.

Exact reproduction guidance and stage-specific caveats remain in the versioned README and protocol files. The code deliberately refuses to overwrite some existing V3 outputs; this protects completed experiments from accidental replacement.

11.5 Verification matrix

Table 11.2: Verification evidence retained by stage.

Stage	Check	Evidence
Stage 02	Complete-window partition	Primary and leftover IDs disjoint; their union equals all 17,419 complete windows; tails separately counted.
Stage 04	Transform and balance	All values finite; 12,270 rows; 4,090 labels each; all 69 people; near-zero CAR and temporal means.
V2	Tensor numerics	TT/TensorLy agreement, orthogonality, full-rank recovery, batch invariance, source transform spot checks.
V2	Model replay	105 classifier pipelines, 60 decompositions, 45 inner and 15 outer splits reproduced.
V3	Input lineage	All V2 windows retained; prior artifact IDs absent; stage-02 arrays fingerprinted unchanged.
V3	Waveform math	Explicit delay contraction equals moment energy; saved predictions and split memberships replayed.
V4	Complete replay	150 model entries, 60 bases, 45 inner and 15 outer splits, scalars, weights, and selections verified.
V4	XGBoost portability	23 UBJ boosters exported/reloaded with predictions matching complete saved pipelines.

11.6 Why a booster file is not the whole model

An exported XGBoost booster receives already prepared features. The clinically meaningful pipeline also includes the training-derived TT/Tucker basis when applicable, feature selection, standardization, class ordering, and subject-level pooling. A booster must therefore be paired with its corresponding complete pipeline and basis provenance. Joblib artifacts should be loaded only from a trusted local source because deserialization can execute code.

11.7 Report provenance

This PDF is compiled from modular LaTeX source under `technical_report`. Numerical tables are transcribed from the versioned JSON reports and cross-checked against the consolidated project record. Figures are native TikZ/PGFPlots diagrams using those same values; no decorative AI image is presented as measured data.

12 Limitations, Threats to Validity, and Future Work

12.1 Internal validity

The project addresses several common leakage routes, but residual threats remain:

- The same public participants were examined across multiple experiment generations. Nested CV protects each run’s inner fitting, but it cannot erase researcher-level adaptation after earlier outer outcomes were seen.
- V1 compares ranks and families without nested selection. Its maximum is exploratory.
- V2 calibration uses 24 windows per training person for computational efficiency. This is reproducible but may underrepresent some within-person variability.
- V3/V4 spectral and waveform branches cover different effective bands: spectra use 1–30 Hz, while waveform moments retain approximately 0.5–45 Hz. The comparison is between complete pipelines, not a perfectly isolated representation ablation.
- Repeated CV folds are correlated. The three repeat scores are too few for a reliable normal-theory confidence interval.

12.2 Construct validity

Balanced accuracy measures classification, not biomarker validity. The project does not establish that any factor corresponds to a neurophysiological disease mechanism. The original proposal’s MMSE correlations, ANOVA of factor loadings, topographic biomarker maps, dPLV connectivity, and Riemannian baselines remain unimplemented. Because channel order was wrong in V1, its graph/electrode interpretations must not be used.

The waveform moment is exact for mean squared linear delay-filter energy, but it cannot represent arbitrary nonlinear waveform dynamics or long-range order. Calling it “raw training” without this qualification would overstate the model.

12.3 External validity

The cohort is single-site, limited to 88 people, and includes only 23 FTD participants. Recording device, clinical workflow, derivative preprocessing, population characteristics, and label criteria may differ elsewhere. No result is calibrated for clinical probabilities, prospective use, or medical decisions.

12.4 Statistical power and uncertainty

The main unit is the person, not the window. This leaves modest test folds of roughly 17–18 people and creates sensitivity to which FTD cases are held out. The 59.7% mean is only about 5.2 percentage points above the V4 primary endpoint and is chosen as a best family after comparison. A formal inferential claim would need an external test cohort or a prespecified locked model, plus uncertainty quantification appropriate for participant-level clustered data.

12.5 Prioritized next experiments

Future work should be sequenced rather than expanded into an unlimited search.

1. **Lock an external-validation protocol.** Freeze one representation, one classifier, all preprocessing thresholds, and a single primary metric before obtaining a new cohort.
2. **Acquire independent FTD participants.** New people are more valuable than duplicating existing windows. Multi-site recruitment would also test device/site generalization.

3. **Run targeted ablations.** Frequency-match waveform and spectral branches; compare TT16, spectral-32, and fusion-32 under identical preprocessing and one fixed classifier.
4. **Add the missing course baselines.** Implement tangent-space Riemannian covariance features, then compare them under the same nested splits.
5. **Investigate connectivity carefully.** Implement dPLV or another phase-connectivity tensor with leakage-safe estimation and surrogate/null checks. Avoid presenting volume-conduction effects as disease connectivity.
6. **Stability and interpretability.** Evaluate Tucker/TT subspace stability across folds, bootstrap electrode/frequency loadings by participant, and correct for multiple comparisons before claiming biomarkers.
7. **Clinically separate objectives.** If MMSE association is studied, treat it as a distinct outcome with its own validation rather than adding MMSE to an EEG diagnosis classifier.

12.6 What should not be done next

The project should not repeatedly expand the same 88-person hyperparameter grid until 65–70% appears. It should not split by window, oversample FTD windows as if they were new patients, merge the leftover directory destructively, or report secondary family maxima as the primary endpoint. Those actions would increase apparent performance more readily than scientific credibility.

13 Conclusion

NeuroTensor developed a complete, versioned course pipeline from public resting-state EEG to tensor representation, low-rank decomposition, participant-level classification, and artifact replay. The work began by solving a practical data-engineering problem: FTD limited both people and duration. An FTD-referenced MILP match produced a controlled 69-person cohort while preserving all excess windows. Signal audit, CAR, deterministic outlier rules, and boundary guards then made the analysis partitions explicit and reproducible.

Three tensor views were tested: a non-negative subject-channel-frequency-time tensor, signed logPSD channel-frequency-local- segment tensors, and raw-waveform channel-lag-position delay tensors summarized by exact second-order moments. CP, graph NCP, Tucker/HOSVD, TT, and supervised TT were evaluated alongside spectral features and standard classifiers.

The highest retained family result is 59.7% mean balanced accuracy for tuned RBF SVM, while the V4 primary complete-pipeline endpoint is 54.5%. CN is the easiest class and FTD remains the hardest. Tucker is competitive in the corrected 69-person experiment, but spectral controls outperform waveform-only methods in the 88-person experiment. Fusion is promising but does not isolate a TT effect. XGBoost is useful yet not best; longer optimization improves neither the fixed supervised TT nor the evidence base by itself.

The project’s strongest contribution is therefore methodological: it shows how to move from attractive tensor ideas to an auditable experiment while preserving negative results and corrections. The current models are meaningful course benchmarks, not clinical systems. The next credible advance is new independent FTD data and a locked external-validation protocol, not another unbounded search on the same participants.

Final takeaway

Tensor decomposition provides a principled language for preserving multi-way EEG structure, but trustworthy performance depends equally on cohort design, subject-level validation, training-only transformations, optimization diagnostics, and honest endpoint reporting.

A Source-Code Function Map

Table A.1: Principal implementation units and responsibilities.

File	Function or class	Role
preprocessing/build_balanced_dataset.py	load_records, load_downsampled	Read BIDS metadata and EEGLAB derivatives; verify sampling and channel count; decimate 500 to 250 Hz.
Same	scan_recordings	Count complete windows and first-pass non-finite, flat, and amplitude flags.
Same	set_reference_buckets	Sort all 23 FTD people and define 8/7/8 duration buckets plus shared thresholds.
Same	match_group	Solve binary MILP assignment with shortage, sex, age, duration mismatch, quota, uniqueness, and bucket-capacity terms.
Same	even_temporal_selection	Choose target windows evenly across each selected recording.
Same	make_entries, write_signal_datasets, verify_outputs	Build shuffled primary/leftover manifests and arrays; preserve partial tails; verify complete partition.
preprocessing/audit_primary_signal_quality.py	band_sum, robust_z, main	Compute robust channel/window/subject QC and spectral diagnostics without mutation.
preprocessing/build_clean_primary_dataset.py	car_transform	Common-average re-reference and per-channel mean removal.
Same	signal_qc_reason, write_partition	Label fixed threshold failures; write clean/excluded arrays and metadata.
tensor_pipeline/tensorize_subjects.py	main	Construct V1 $69 \times 19 \times 59 \times 10$ non-negative spectral tensor.
tensor_pipeline/run_decomposition_benchmark.py	fit_nonnegative_cp_mu, decompose	Fit PCA/NMF/CP/graph-CP/Tucker on training folds and project held-out subjects.
Same	classifier, aggregate_metrics	Build standardized classifiers and subject-level multiclass metrics.
tensor_pipeline/diagnose_feature_bottleneck.py	representations, main	Compare less-compressed channel-band-time features with training-only selection.
tensor_pipeline/v2/prepare.py	main	Verify header order, apply boundary guards, create logPSD tensors and 418 controls.
tensor_pipeline/v2/models.py	TTPProjection	Covariance-based TT-SVD-equivalent common basis with reusable held-out transform.
Same	TuckerProjection	Training-only HOSVD basis for signed log-power tensors.
Same	anatomical_smoothing	Four-neighbor coordinate graph and normalized-Laplacian smoother.
tensor_pipeline/v2/benchmark.py	Experiment.features	Fit/cache training-only decomposition bases and pool window latent features.
Same	Experiment.tune, outer, summarize	Nested subject CV, deterministic selection, persistence, and repeat aggregation.
tensor_pipeline/v3_waveform/prepare.py	delay_moment	Build $19 \times 16 \times 485$ delay view and exact 304×304 covariance moment.

File	Function or class	Role
Same	spectral_features, main	Recompute 418 controls on full cohort and verify lineage/exclusions.
tensor_pipeline/v3_waveform/wave_models.py	weights, balanced_moment, energies	Equalize class loss, build training moments, and evaluate normalized filter energy.
Same	DelayProjection	Fit unsupervised waveform TT or Tucker filters.
Same	SupervisedTT	Jointly optimize TT cores and a three-class softmax head with analytic gradients.
Same	WavePipeline	Combine representation, selection, weighted scaling, and classifier.
tensor_pipeline/v3_waveform/run.py	main, summarize	Frozen 26-config nested benchmark and fixed learning-curve diagnostic.
tensor_pipeline/v3_waveform/convergence_check.py	main	Post-hoc 100 versus 2,000 iteration sensitivity with identical splits.
tensor_pipeline/v4_tuning/tune_models.py	RegularizedTT	Replace weak core anchor with configurable 0.1/1.0 anchoring.
Same	feature_bank, classifier_specs, configs	Fit training-only TT/Tucker banks and enumerate exactly 139 settings.
tensor_pipeline/v4_tuning/benchmark_tuning.py	rank_rows, select	Deterministic primary, classifier-family, and representation-family inner selection.
Same	main	Fit nested V4 benchmark, save bases/models/predictions, and aggregate complete repeats.
tensor_pipeline/v4_tuning/verify_report.py	main	Replay 150 entries, reproduce split/scaler/weight/selection behavior, and export/reload XGBoost boosters.

B Hyperparameter and Representation Reference

Table B.1: Key fixed and candidate hyperparameters.

Stage	Component	Values
All spectral	Welch/spectrogram	250 Hz; Hann; 500 samples (2 s); 250 overlap; 500 FFT; 0.5 Hz bins; primary 1–30 Hz.
V1	Tensor	69 subjects $\times$ 19 channels $\times$ 59 frequencies $\times$ 10 rank-time bins; square-root relative power; median pooling.
V1	Ranks	3, 5, 8 for PCA/NMF/CP/graph CP/Tucker comparison.
V1	NMF	NNDSVDa; coordinate descent; 1,000 iterations; tolerance 10^{-5} ; rank-8 sensitivity at 5,000.
V1	Graph CP	Nominal graph $\lambda = 0.1$; preliminary due objective-scaling caveat.
V2	Window tensor	$19 \times 59 \times 3$ signed log10 PSD; subject statistics mean/median/SD.

Stage	Component	Values
V2	TT	terminal ranks 8, 16, 32; default channel rank 8; spectral rank 24; 24 calibration windows per training person.
V2	Tucker	ranks (6, 8, 3); selected latent counts 16, 32, 64.
V2	Classifiers	LR $C = 0.1, 1, 10$; linear SVM $C = 0.1, 1$; RBF SVM $C = 1, 10$; fixed HGB.
V3/V4	Delay tensor	125 Hz; 19 channels; 16 lags; 485 local positions; 120 ms span; subject moment 304×304 .
V3	Wave TT/Tucker	TT channel rank 6, terminal 8/16; Tucker ranks 4/6; LR $C = 0.1, 1$; RBF SVM $C = 1, 10$.
V3	Supervised TT	channel rank 4; filters 8; head penalty 0.01/0.1; core anchor 0.001; 100 iterations.
V4	Logistic	$C = 0.01, 0.1, 1$; max 5,000 iterations; tolerance 10^{-5} .
V4	RBF SVM	$C = 0.1, 1, 10$; $\gamma = \text{scale or } 0.01$; uncalibrated decision function.
V4	Random forest	200 trees; depth 3/min leaf 3 or unlimited/min leaf 5; square-root features; seed 9064.
V4	XGBoost	Eight fixed combinations: depth 1–3; rate 0.03–0.1; 100–250 trees; child 1–5; L_2 1–10; L_1 0–0.5; row 0.8–1; columns 0.8.
V4	Supervised TT	six fixed settings; channel rank 2/4; filters 4/8; head penalty 0.1/1; anchor 0.1/1; 500 iterations.
V2	CV	Outer 3 repetitions $\times$ 5 folds, seed 5040; inner 3 folds, seed 8000+ outer index.
V3/V4	CV	Outer 3 repetitions $\times$ 5 folds, seed 9060; inner 3 folds, seed 90600+ outer index.

B.1 Exact V4 XGBoost tuples

Each tuple is ordered as (depth, learning rate, trees, minimum child weight, L_2 , L_1 , row subsample), with column subsample fixed at 0.8:

1. (1, 0.03, 250, 1, 10, 0.1, 1.0)
2. (1, 0.10, 100, 3, 10, 0.1, 0.8)
3. (2, 0.03, 250, 1, 1, 0, 1.0)
4. (2, 0.10, 100, 3, 10, 0.1, 0.8)
5. (2, 0.03, 150, 5, 10, 0.5, 1.0)
6. (3, 0.03, 250, 3, 10, 0.5, 0.8)
7. (2, 0.10, 100, 1, 1, 0.1, 0.8)
8. (3, 0.10, 100, 5, 10, 0.5, 1.0)

C Reports, Data Partitions, and Model Artifacts

Table C.1: Canonical project evidence map.

Path	Purpose
PROJECT_MASTER_REPORT.md	Consolidated project record through V4; starting point for chronology and cross-stage caveats.
preprocessing/reports/00_CURRENT_INDEX.md	Current preprocessing report index; identifies older stale narrative pages.
preprocessing/02_balanced_primary/processing_report.json	Source counts, bucket thresholds, matching demographics, primary/leftover statistics, and partition checks.
preprocessing/03_signal_qc_audit/audit_report.json	Signal-quality summaries and outlier diagnostics.
preprocessing/04_clean_primary/processing_report.json	CAR, final thresholds, exclusion reasons, balance trims, and clean-primary verification.
tensor_pipeline/outputs/summary/summary.json	V1 rank sweep, metrics, confusion matrices, and reconstruction diagnostics.
tensor_pipeline/outputs/summary/V2_AUDIT_NOTE.md	Channel-order and preliminary graph-objective correction note.
tensor_pipeline/v2/AUDIT_AND_PROTOCOL.md	V2 methodological protocol and leakage controls.
tensor_pipeline/outputs/v2_tensor_train/	V2 tensors, controls, indexes, bases, fold models, nested metrics, manifest, and verification.
tensor_pipeline/v3_waveform/PROTOCOL.md	Full-cohort waveform tensor, class/subject weighting, and learning-curve protocol.
tensor_pipeline/outputs/v3_waveform_full_cohort/	V3 moments, controls, indexes, 15 fold outputs, saved models, convergence sensitivity, and verification.
tensor_pipeline/v4_tuning/PROTOCOL.md	Frozen V4 representation/classifier/regularization grid and endpoint definitions.
tensor_pipeline/outputs/v4_tuning/protocol.json	Exact 139 configurations and shared V3 input signature.
tensor_pipeline/outputs/v4_tuning/nested_report.json	Primary, classifier-family, and representation-family V4 metrics.
tensor_pipeline/outputs/v4_tuning/selected_configurations.json	Per-outer-fold selected settings.
tensor_pipeline/outputs/v4_tuning/verification_report.json	PASS record for 150 models, 60 bases, 60 splits, weights, scalers, selections, and XGBoost exports.
technical_report/	Modular LaTeX source, figures, appendices, bibliography, and build guide for this report.

C.1 Canonical versus historical material

The canonical controlled data are under `preprocessing/04_clean_primary/primary`. The V3/V4 full-cohort analysis uses a separate index over stage-02 primary and leftover arrays; it does not replace

the controlled partition. The original `preprocessing/reports/README.md` and `tensor_pipeline/README.md` are historical snapshots and include statements that became stale after later experiments. The `00_CURRENT_INDEX.md` files and this report provide current navigation.

D Glossary and Notation

AD Alzheimer’s disease.

ASR Artifact Subspace Reconstruction, applied by the dataset authors to supplied derivatives.

Balanced accuracy (BA) Arithmetic mean of per-class recalls; primary metric for three-class comparisons.

Boundary guard Exclusion of any four-second window intersecting an EEGLAB boundary expanded by 0.5 s on each side.

CAR Common-average reference across the 19 scalp channels at each time sample.

CN Cognitively normal control group.

CP CANDECOMP/PARAFAC decomposition: sum of tensor rank-one terms.

FTD Frontotemporal dementia.

HOSVD Higher-order singular value decomposition; used here as a training-only Tucker/multilinear projection.

Inner CV Cross-validation conducted only inside outer training people to choose ranks, features, and classifiers.

Outer CV Held-out participant evaluation of the complete selection and fitting procedure.

PSD Power spectral density.

Subject pooling Reduction of multiple windows to one equal-weight participant representation.

Tensor Train (TT) Chain of low-rank cores whose matrix products recover tensor entries.

Tucker decomposition Multilinear core tensor multiplied by one factor matrix per mode.

Window One non-overlapping four-second, 19-channel segment at 250 Hz in the processed arrays.

D.1 Principal symbols

Symbol	Meaning
X	General multi-way tensor
R	CP or terminal latent rank
U, V, G	Factor matrices or TT/Tucker cores
A	Electrode adjacency matrix
L	Electrode graph Laplacian
S	Subject delay-covariance moment
b	Vectorized channel-lag filter
ω_i	Training subject class weight
M	Confusion matrix

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